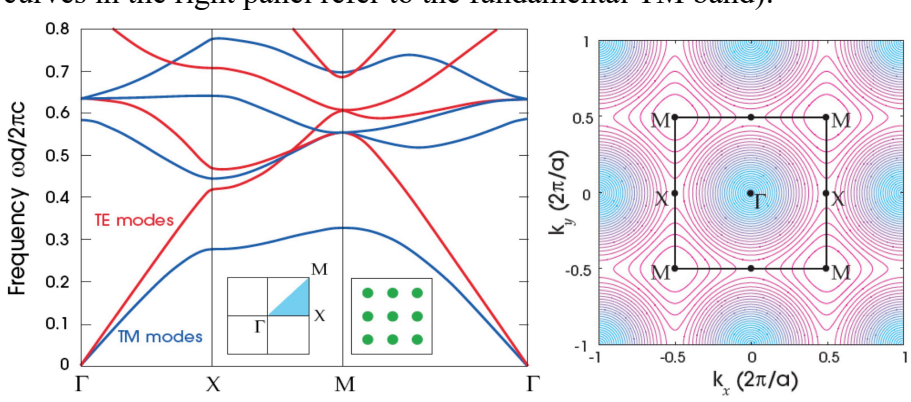
Some examples of exam questions

THEORETICAL FOUNDATIONS

1. (i) Write the time-harmonic Maxwell's equations and constitutive relations in a linear dielectric medium and (ii) derive, under suitable hypotheses, the Master Equation (i.e. eigenvalue equation) for the harmonic modes in terms of the magnetic field.
2. Define (i) the scalar product and (ii) Hermiticity in electromagnetics. (iii) Define the L_H operator for electromagnetic harmonic modes in normal media. (iv) Demonstrate that the frequency ω of any harmonic mode of a normal medium is real valued, and (v) harmonic modes of different frequency are orthogonal.
3. Provide some examples of symmetry in an optical medium and subsequent properties of the harmonic modes.
4. Demonstrate the scale invariance of the Master Equation of harmonic modes for a (source-free) linear, lossless and non-dispersive medium.
5. Discuss the main properties of evanescent waves at the interface between two dielectric structures.

PHOTONIC CRYSTALS

6. (i) State the Bloch Theorem in Electromagnetics for a generic photonic crystal.
   (ii) Summarizes the procedure to derive the eigenvalue equation of a periodic layered medium (PLM).
   (iii) From the eigenvalue equation, derive the dispersion relation of the harmonic modes of a PLM and comment on the solutions.
7. Consider an interface between air and a periodic layered medium (PLM) made of normal media: (i) State the ansatz of solution for Surfaces States (SS) harmonic modes; (ii) Summarizes the procedure to derive the dispersion law of the fundamental SS; (iii) Sketch in the w-k plane the dispersion diagram of SS harmonic modes.
8. Consider a 2D photonic crystal with the following band structure (the isofrequency curves in the right panel refer to the fundamental TM band):



Describe and explain the origin of negative refraction and superprism effects (in the xy plane) taking place at the flat interface between the photonic crystal and a homogeneous medium (e.g. air).

NEAR-FIELD OPTICS AND PLASMONICS

9. Derive the Drude permittivity formula for noble metals and comments on the results.
10. Explain the origin of the diffraction limit to conventional (i.e. far-field) microscopy and then explain the principle of operation of a near-field microscope.
11. Derive the condition of existence of Surface Plasmon Polariton waves propagating at the flat interface between two homogeneous (lossless) media.
12. Write and sketch (in the proper plane) the dispersion relation of surface-plasmon polaritons and describe their properties.
13. Derive the expression of the polarizability of a metal nanosphere embedded in a homogeneous dielectric medium under the quasi-static approximation.
14. What are the optical properties of a single metal nanosphere embedded in a homogeneous dielectric, under the quasi-static approximation?

METAMATERIALS

15. Demonstrate that an array of metallic wires embedded in a homogeneous dielectric can exhibit a negative effective permittivity in a suitable range of frequencies.
16. Demonstrate that an array of metallic split-cylinders embedded in a homogeneous dielectric can exhibit a negative effective permeability in a suitable range of frequencies.
17. Derive the condition of existence of Surface Magnon Polariton waves propagating at the flat interface between two (effectively) homogeneous (lossless) media (metamaterials).
18. Demonstrate that for a double negative medium (both negative permittivity and permeability, real parts) is a left-hand medium and ought to exhibit a negative refractive index, real part.